

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO3_AT3 — VISUALIZATION CRITIQUE & REDESIGN

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO3 – Evaluation (BL5)	Assessment:	Assessment Tool 3 – Critique & Redesign
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO3 Statement:	<i>Evaluate flawed or misleading data visualizations, critique their specific shortcomings, and justify an improved redesign.(BL5)</i>		
Student Name:	P.Vishwa Nandan	Reg. No.:	192473015
Section / Batch:	2024	Date:	18-08-2026

Code of Conduct

/ P.Vishwa nandan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: vishwa

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond simply naming a flaw: explain WHY it is misleading and what a reader could wrongly conclude from it.
- Your redesign must be a specific, justified chart proposal — not just “use a different chart.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – VISUALIZATION CRITIQUE & REDESIGN

Rubrics for Calculation **ASSESSMENT RUBRIC**
AT3 — Visualization Critique & Redesign
CO3 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO3-AT3 Visualization Critique & Redesign problems, in which students are given a flawed or misleading chart, must critique its specific shortcomings, and propose a justified redesign.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the specific visualization principle being violated (e.g., proportional ink, axis scaling, color perception, overplotting) in the given flawed chart.	Good understanding of the relevant principle, with only minor gaps in identifying exactly what is violated.	Basic understanding, but with some misconceptions about which principle applies to the flaw shown.	Poor understanding of core principles; misidentifies the nature of the flaw entirely.
Explanation	25%	Clearly explains why the flaw is misleading, including what a reader could wrongly conclude as a result — not just that something is 'wrong.'	Explains the flaw with adequate reasoning, covering most of why it misleads a reader.	Identifies the flaw but explanation is generic or only loosely tied to its misleading effect.	Little or no explanation of why the flaw is misleading; the critique is a bare assertion.
Application	20%	Proposes a specific, well-justified redesign that directly resolves the identified flaw,	Proposes a workable redesign that resolves the flaw, with only minor gaps in	Proposes a redesign, but it only partially resolves the flaw or lacks sufficient justification.	Fails to propose a workable redesign, or the redesign does not address the identified flaw.

		correctly applying appropriate chart-selection and design principles.	justification.		
Organization	15%	The response is clearly structured, addressing both the critique and the redesign in a logical sequence with coherent overall flow.	The response is organized and addresses both parts, with only minor issues in flow or clarity.	Basic structure; both parts are addressed but the response lacks clarity or a logical sequence.	Poorly organized; the response is incoherent, and one or both parts are left unaddressed.
Accuracy	10%	Both the critique and the redesign are completely accurate and well-suited to the specific chart and data described.	Mostly accurate, with only minor omissions or a small error in the critique or redesign.	Partially correct; the critique or redesign contains notable inaccuracies or mismatches to the scenario.	Largely incorrect or inappropriate, with major gaps in both the critique and the redesign.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 2 — 3D Pie Chart

a) Critique

A 3D tilted pie chart introduces visual distortion because the slices closer to the viewer appear larger than slices farther away. The tilt changes the apparent angles and areas of the slices, making it difficult to accurately compare the revenue shares.

Therefore, a region with a smaller actual percentage may look larger because of its position in the tilted chart.

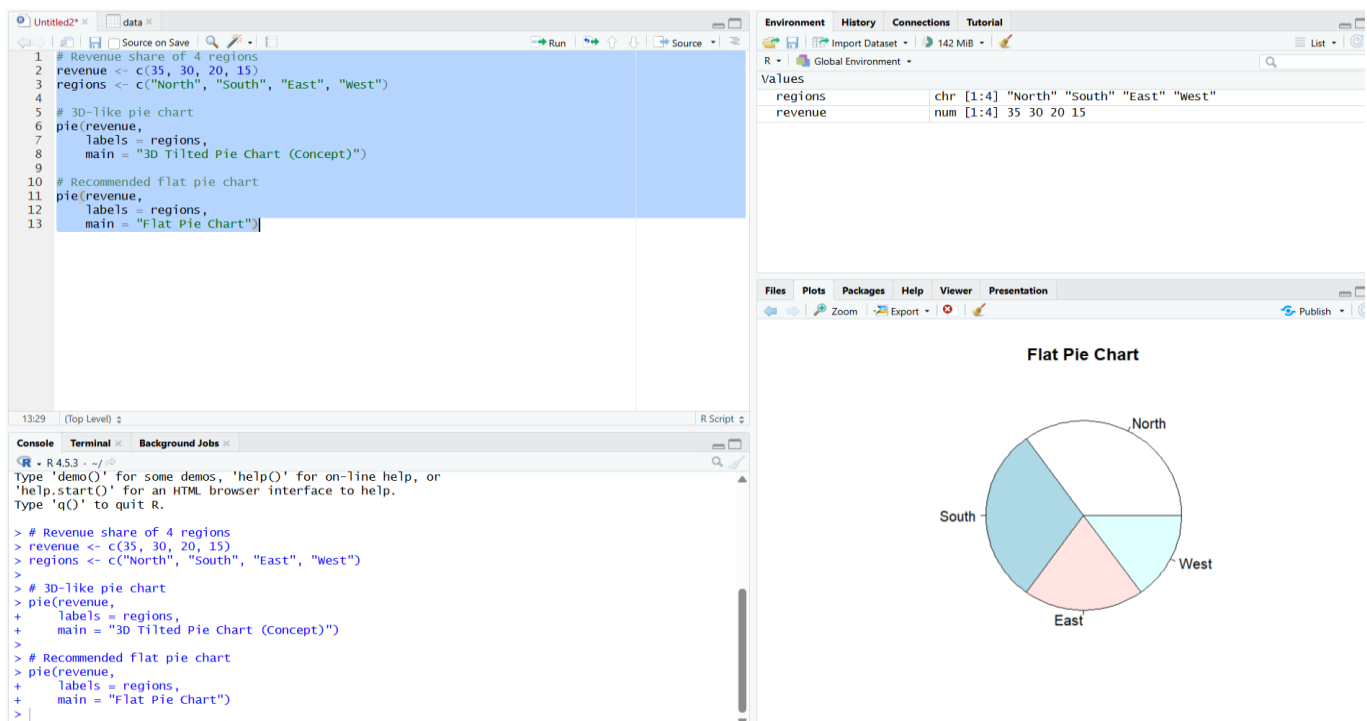
Code:

```
# Revenue share of 4 regions
revenue <- c(35, 30, 20, 15)
regions <- c("North", "South", "East", "West")
```

```
# 3D-like pie chart
pie(revenue,
    labels = regions,
    main = "3D Tilted Pie Chart (Concept)")
```

```
# Recommended flat pie chart
pie(revenue,
    labels = regions,
    main = "Flat Pie Chart")
```

Output:



b) Redesign

Use a flat 2D pie chart or, preferably, a bar chart.

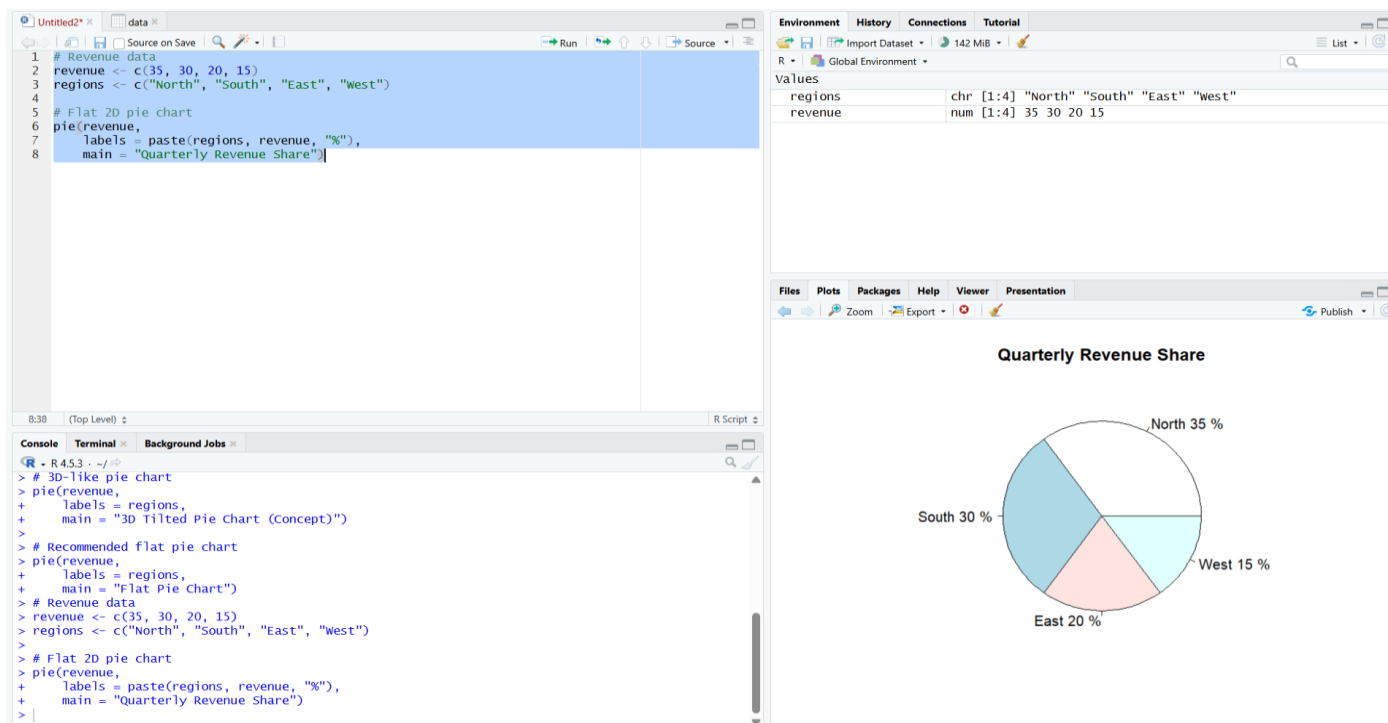
A flat pie chart removes the 3D perspective and represents each slice using its true angle and area, making comparisons more accurate.

Code:

Flat alternative

```
pie(revenue,  
    labels = paste(regions, revenue, "%"),  
    main = "Quarterly Revenue Share")
```

Output:



Problem 8 — Stacked Bar Charts

a) Critique

Using different department orders and colors in the 2022 and 2023 stacked bar charts makes year-over-year comparison difficult and confusing.

For example, if the Sales department is blue in 2022 but green in 2023, the reader may incorrectly think they represent different departments. Similarly, changing the order of departments makes it difficult to compare the size of the same segment across the two years.

This inconsistency increases the chance of misinterpretation and visual errors.

b) Redesign

For a fair comparison, use the same department order and the same colors in both 2022 and 2023 charts.

For example:

- Sales → Blue
- HR → Orange
- Finance → Green
- IT → Red

The departments should appear in the same order from bottom to top in both stacked bars.

Code:

Data

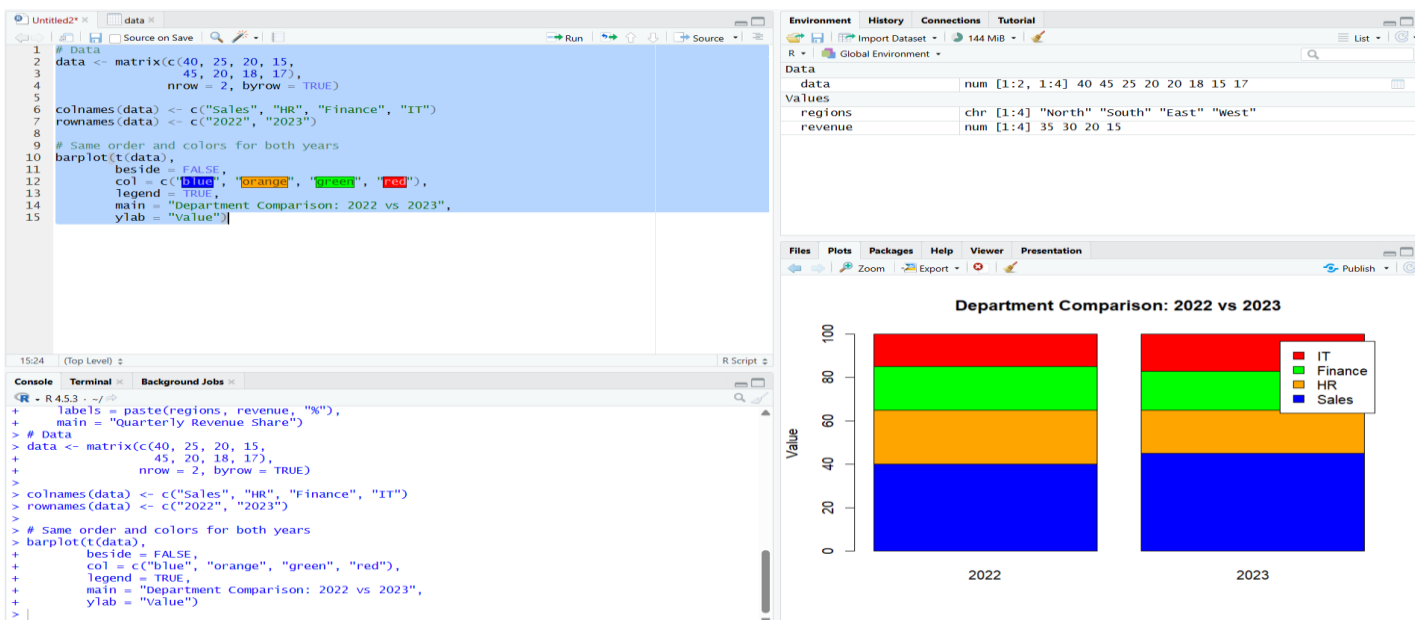
```
data <- matrix(c(40, 25, 20, 15,  
                45, 20, 18, 17),  
              nrow = 2, byrow = TRUE)
```

```
colnames(data) <- c("Sales", "HR", "Finance", "IT")  
rownames(data) <- c("2022", "2023")
```

Same order and colors for both years

```
barplot(t(data),  
        beside = FALSE,  
        col = c("blue", "orange", "green", "red"),  
        legend = TRUE,  
        main = "Department Comparison: 2022 vs 2023",  
        ylab = "Value")
```

Output:



Problem 14 — Histogram Bin Width

a) Critique the bin-width choice

Using only 2 very wide bins is not appropriate for an exam-score histogram. It combines too many observations into each bin and hides important features of the data, such as:

- The shape of the distribution
- Peaks or modes
- Skewness
- Gaps or clusters
- Possible outliers

As a result, the reader cannot properly understand how the exam scores are distributed.

b) Redesign

Use a smaller and suitable bin width so that the distribution can be seen clearly. The bin width can be selected using methods such as Freedman–Diaconis or Sturges' rule.

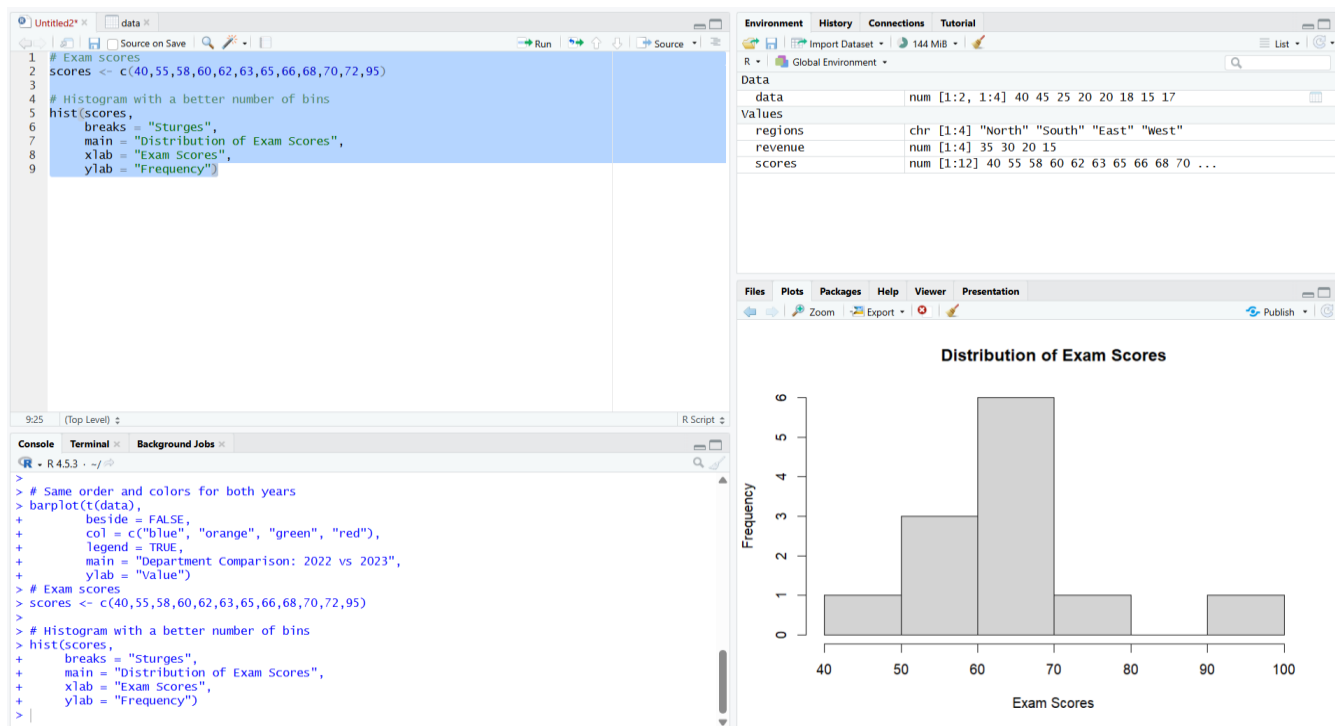
In R, you can let R choose a reasonable number of bins:

Code:

```
# Exam scores
scores <- c(40,55,58,60,62,63,65,66,68,70,72,95)

# Histogram with a better number of bins
hist(scores,
      breaks = "Sturges",
      main = "Distribution of Exam Scores",
      xlab = "Exam Scores",
      ylab = "Frequency")
```

Output:



Problem 20 — Misleading Pie Charts

a) Critique

Drawing two pie charts with the same circle size can give a misleading impression when their total values are very different.

For example, if Employee A has a total performance value of ₹100,000 and Employee B has ₹20,000, but both pie charts are drawn with the same visual size, the reader may think that their overall performance is similar.

A pie chart with a fixed size mainly shows proportions, not the difference in total magnitude. Therefore, the large difference between the employees can be hidden.

b) Redesign

A better approach is to use a bar chart showing the actual total performance values. This makes the difference in magnitude immediately visible.

Code:

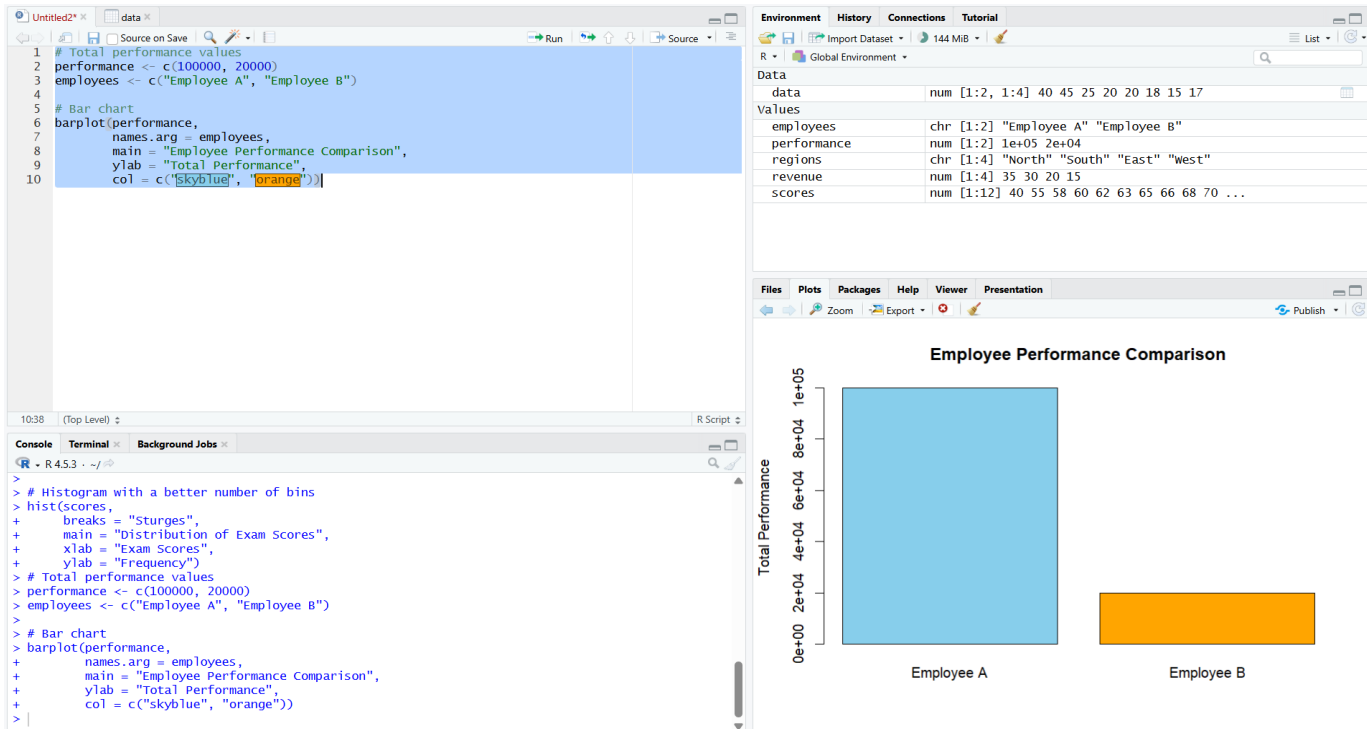
```
# Total performance values
performance <- c(100000, 20000)
employees <- c("Employee A", "Employee B")
```

```
# Bar chart
barplot(performance,
```



```
names.arg = employees,
main = "Employee Performance Comparison",
ylab = "Total Performance",
col = c("skyblue", "orange"))
```

Output:



Problem 21 — Misleading Hierarchical/Flow Charts

a) Critique

A treemap is not ideal for this situation because treemaps are mainly designed to represent hierarchical data using nested rectangles.

Here, the budget has only three simple categories and there is no hierarchy. Using multiple nested rectangles adds unnecessary visual complexity and makes the chart harder to read.

It may also make the viewer spend time understanding the structure instead of simply comparing the three budget categories.

b) Redesign

A simple bar chart is a better choice because it clearly compares the budget amount for each category.

Code:

```
# Budget data
budget <- c(50000, 30000, 20000)
category <- c("Education", "Infrastructure", "Other")
```

```
# Bar chart
barplot(budget,
        names.arg = category,
        main = "Budget Breakdown",
        xlab = "Category",
        ylab = "Budget Amount (₹)",
        col = "skyblue")
```

Output:

